

EFFECT OF INSTANTANEOUS PERFORMANCE FEEDBACK DURING 6 WEEKS OF VELOCITY-BASED RESISTANCE TRAINING ON SPORT-SPECIFIC PERFORMANCE TESTS

AARON D. RANDELL,¹ JOHN B. CRONIN,^{1,2} JUSTIN W. L. KEOGH,¹ NICHOLAS D. GILL,¹
AND MURRAY C. PEDERSEN²

¹*Sport Performance Research Institute New Zealand, AUT University, Auckland, New Zealand;* ²*School of Exercise, Biomedical and Health Sciences, Edith Cowan University, Joondalup, Western Australia, Australia; and* ³*Bay of Plenty Rugby Union, Bay Park Stadium, Mount Maunganui, New Zealand*

ABSTRACT

Randell, AD, Cronin, JB, Keogh, JW, Gill, ND, and Pedersen, MC. Effect of instantaneous performance feedback during 6 weeks of velocity-based resistance training on sport-specific performance tests. *J Strength Cond Res* 25(1): 87–93, 2011—The purpose of this study was to investigate the effect of instantaneous performance feedback (peak velocity) provided after each repetition of squat jump exercises over a 6-week training block on sport-specific performance tests. Thirteen professional rugby players were randomly assigned to 1 of 2 groups, feedback ($n = 7$) and non-feedback ($n = 6$). Both groups completed a 6-week training program (3 sessions per week) comprising exercises typical of their normal preseason conditioning program. Squat jumps were performed in 2 of the 3 sessions each week during which both groups performed 3 sets of 3 concentric squat jumps using a barbell with an absolute load of 40 kg. Participants in group 1 were given real-time feedback on peak velocity of the squat jump at the completion of each repetition using a linear position transducer and customized software, whereas those in group 2 did not receive any feedback. Pre and posttesting consisted of vertical jump, horizontal jump, and 10-/20-/30-m timed sprints. The relative magnitude (effect size) of the training effects for all performance tests was found to be small (0.18–0.28), except for the 30-m sprint performance, which was moderate (0.46). The probabilities that the use of feedback during squat jump training for 6 weeks was beneficial to increasing performance of sport-specific tests was 45% for vertical jump, 65% for 10-m sprints, 49% for 20-m sprints, 83% for horizontal jump, and

99% for 30-m sprints. In addition to improvements in the performance of sport-specific tests, suggesting the potential for greater adaptation and larger training effects, the provision of feedback may also be used in applications around performance targets and thresholds during training.

KEY WORDS measurement technology, jump squats, power, transference

INTRODUCTION

Current monitoring practices typically provide retrospective quantification of a resistance training session. That is, the information collected summarizes a completed session and is therefore used to modify a subsequent session. It is fairly conclusive from motor learning theory, however, that instantaneous feedback in terms of knowledge of performance and knowledge of results can have a substantial effect on athletic performance and the acquisition of motor skills (2,12). Of particular interest is the literature citing improvements in strength and the acute production of force and power (6–12% improvements) when the subjects were exposed to visual feedback (5,7,11). However, the effects of this type of feedback over an entire resistance strength training cycle are unexplored and provide exciting possibilities for improved athletic performance.

Advances in technology (linear position transducers, rotary encoders, etc.) now enable the direct measurement of many kinematic (e.g., velocity) and kinetic (e.g., power) variables during certain resistance training exercises. Although this type of data is used effectively to test the effects of resistance training through assessments, its major benefit may be the ability to continuously monitor performance during training (4). Although the monitoring of training load and or training intensity may provide useful information as to what has been completed its value in affecting positive changes within a session or to quantify and evaluate each session is limited. A natural progression would be to constantly monitor each

Address correspondence to Aaron D. Randell, aaron.randell@aut.ac.nz.
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Figure 1. Setup of barbell, adjustable rack, and linear position transducer for performance of squat jumps.

training session and offer specific, individualized feedback provided by these recent advances in technology, which may result in superior performance gains than a session in which no feedback was given. In other words, to ensure an optimal training stimulus for adaptation, it is hypothesized that feedback should be provided after each repetition over the entire duration of the training session. Currently, there is a paucity of research in this area.

What is apparent from the literature is that the strength endurance and strength phases of the training pyramid are adequately quantified via load, intensity, and volume.

However, the ability to relevantly quantify and provide feedback on the power phase remains relatively unexplored and requires future investigation. Although the load or the intensity of the load lifted appears to be an important variable to consider for strength endurance and strength adaptation, other variables could possibly be of greater importance for power adaptation. That is, how the load is actually moved may be more significant in developing and explaining improvements to functional performance (8,9,13). Maximum power output is the product of optimum force and optimum shortening velocity (6,20); therefore, when training for power development, it would seem intuitive to ensure movement velocity and/or force output or power output for each repetition of an exercise session is maximized. Consequently, it would seem logical to monitor and provide feedback for these variables. It is hypothesized that repetition by repetition feedback on bar velocity may enhance the development of power. Therefore, the purpose of this study was to investigate the effect of instantaneous performance feedback (peak velocity) provided after each repetition of squat jump exercises over a 6-week training block on sport-specific performance tests.

METHODS

Experimental Approach to the Problem

A randomized control training study of 6-week duration was used to determine the effect of a feedback or non-feedback squat jump intervention on functional performance. Thirteen subjects were randomly assigned to a feedback or non-feedback group. The bar velocity during squat jumps was

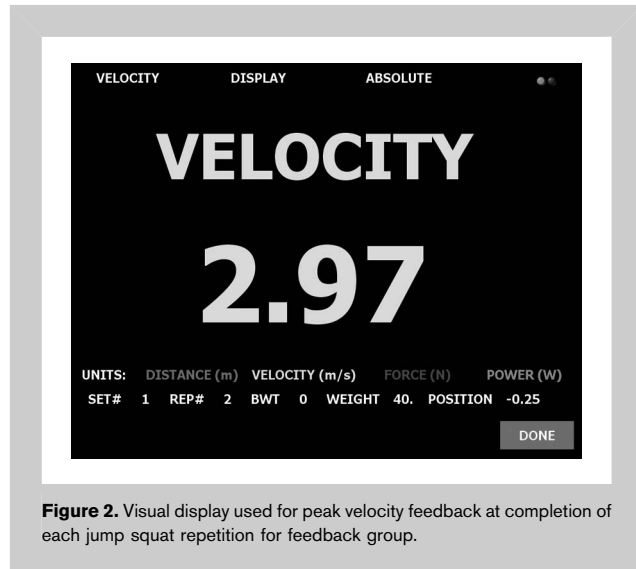
TABLE 1. Six-week preseason resistance training program.*†

Monday	Wednesday	Friday
Shoulder prehab 4 × 20	Squat jumps 3 × 3 (40 kg)	Squat jumps 3 × 3 (40 kg)
Bent over row + pull up 3 × 6 RM	Deadlift + front squat + push press 4 × 3 RM	Deadlift + front squat + push press 5 × 6 RM
Or lat pull over 3 × 6		
Bench press 5 × 6 RM	Deep squat 4 × 6 RM	Bench press 5 × (6–1) RM‡
Deep squat + shoulder press 3 × 6 RM	Calf raise 4 × 6 RM	Bent over row 5 × 6 RM
Or DB deadlift + shoulder Press 3 × 6 RM		
Or shoulder press + rotation 3 × 6 RM		
Single arm bench row 3 × 6 RM	Single leg press 4 × 6 RM	Deadlift + shrug + upright row 5 × 6 RM
Or bent over row 3 × 6 RM		
Side bend 3 × 5 RM	Glute ham raise 4 × 6 RM	Shoulder press 3 × 6 RM
Or decline bench press 5 × 6 RM	Or leg curl 4 × 6 RM	Or cable rotation 3 × 6 RM
Or triceps extension 3 × 10 RM		

*RM = repetition maximum.

†All exercises were performed with 2-minute rest between sets.

‡Weight increased each set, that is, from 6RM to 5RM to 4 RM, etc.



quantified for each training session with a linear position transducer. Given that power is the product of velocity and force, it is suggested that maximizing the velocity of the movement may enhance the development of power if force remains unaffected. Differences pre to posttraining in sport-specific performance tests and chances (% and qualitative) that the true value of the statistic was practically or mechanistically positive, trivial, or negative were calculated.

Subjects

For the period of the study, 13 professional rugby players were randomly assigned to 1 of 2 groups, feedback ($n = 7$, age = 25.7 ± 3.6 years, height = 188.5 ± 8.2 cm, weight = 104.3 ± 10.0 kg, training age = 3.7 ± 1.0 years, 1RM squat = 176.0 ± 35.6 kg) and non-feedback ($n = 6$, age = 24.2 ± 2.5 years, height = 184.7 ± 7.2 cm, weight = 102.9 ± 14.3 kg, training age = 3.2 ± 1.2 years, 1RM squat = 185.4 ± 28.8 kg). All subjects had a minimum of 2 years' resistance training experience and were currently in the preseason phase of their training program. All testing procedures and risks were fully explained, and participants were asked to provide their written consent before the start of the study. The study was approved by the AUT University Ethics Committee.

Equipment

A wire from a linear position transducer (Celesco PT5A-150; Chatsworth, CA, USA) was attached to an Olympic barbell. The barbell was loaded with 2 10-kg plates for an absolute load of 40 kg. The barbell was placed on an adjustable squat rack, which was adjusted to the appropriate depth relating to the height of each individual (Figure 1). A Vertec (Swift Performance Equipment, Lismore, Australia) was used to measure vertical jump height. Wireless timing lights (Brower Timing Systems LLC, Draper, UT, USA) set at a height of 90 cm were used to record sprint times over 10/20/30 m.

Procedures

Participants were matched by playing position and randomly assigned to 1 of 2 groups with each group completing a testing sessions at least 48 hours before the commencement of the training study and 48 hours after the completion of training. The testing session was a series of performance tests that the participants completed on a regular basis as part of their conditioning program, so familiarization was unnecessary. A standardized warm-up was undertaken before each testing occasion, which was also performed regularly by the participants. Each testing session consisted of a vertical jump, horizontal jump, and 30-m timed sprints with split times also taken at 10 and 20 m.

Vertical Jump. Subjects stood with both feet on the ground shoulder width apart, and the maximum vertical reach of a single arm was recorded on the Vertec. A countermovement vertical jump was performed, and the maximal reach of the same arm was recorded. The difference between the jumping reach and the standing reach was recorded as the jump height. A minimum of 1-minute rest was given between trials. The better of 2 attempts was used for analysis.

Horizontal Jump. Subjects stood with feet shoulder width apart with toes behind (touching) a line on the ground. Subjects then performed a countermovement horizontal jump, with arm swing allowed, along the length of a tape measure secured to ground. The landing placement of the feet was recorded and the distance from the heel of the foot back to the start line was recorded as the jump distance. If the subjects landed with 1 foot ahead of the other, the jump was not recorded. The better of 2 successful attempts was recorded, and a minimum of 1-minute rest was given between trials.

Timed Sprints. Subjects completed 2 trials of a 30-m maximal sprint with split times also recorded at 10 and 20 m. Times were recorded using a series of wireless timing lights. Subjects self-started from a stationary split stance start with the front of the leading foot 50 cm back from the first timing light. The better of 2 trials (based on 30-m time) was recorded, and a minimum of 2 minutes' rest was given between trials.

Training Program. The exercises and sessions prescribed were part of the regular preseason training program used by the team (see Table 1). Other conditioning sessions involved a energetic and skills focus; however, these sessions were similar for all players. During each session, all participants completed the same number of repetitions, which was adjusted depending on the exercise (Table 1). The subjects in group 1 (feedback) were given real-time feedback (visual onto a screen) on peak velocity at the completion of each repetition (Figure 2), whereas those in group 2 (non-feedback) did not receive any feedback. Subjects performed 3 sets of 3 concentric squat jumps using a barbell with an absolute load of 40 kg. The depth of the squat was set at a knee angle

TABLE 2. Mean (*SD*) values and percent change in mean of vertical jump (m), horizontal jump (m), and 10-/20-/30-m sprints (seconds) pre and post 6-week squat jump training.

	Vertical jump			Horizontal jump			10-m Sprint			20-m Sprint			30-m Sprint		
	Pre	Post	Percent change	Pre	Post	Percent change	Pre	Post	Percent change	Pre	Post	Percent change	Pre	Post	Percent change
Feedback	0.61 (0.06)	0.64 (0.07)	4.6	2.50 (0.16)	2.56 (0.15)	2.6	1.74 (0.04)	1.73 (0.05)	1.3	3.03 (0.06)	3.00 (0.06)	0.9	4.20 (0.11)	4.14 (0.11)	1.4
Non-feedback	0.66 (0.06)	0.67 (0.01)	2.8	2.58 (0.20)	2.59 (0.20)	0.5	1.79 (0.10)	1.79 (0.09)	0.1	3.06 (0.16)	3.06 (0.15)	0.1	4.25 (0.21)	4.26 (0.19)	-0.3

of 90°, and this was controlled using an adjustable rack that the barbell had to rest on before the commencement of each repetition. Participants were instructed to perform the movement as fast and explosively as possible with a pause between repetitions to distinguish each movement.

Statistical Analyses

Peak velocity during the concentric phase for each repetition was recorded using a position transducer with a velocity repeatability of better than $\pm 0.10\%$ of output, and customized data acquisition and analysis software (Labview, National Instruments, Austin, TX, USA). Velocity was differentiated from the displacement time data, which was sampled at 500 Hz and low-pass filtered at 10 Hz.

Intraclass correlation coefficients (ICCs) were used to determine the consistency of effort (i.e., consistency of session average peak velocity) for both groups over the entire training study. A spreadsheet for analysis of a straight forward controlled trial (10) was used to determine the percent change between pre and posttraining study for each of the variables of interest (vertical jump height, horizontal jump distance, and 10-/20-/30-m sprint times). Cohen effect sizes (ES) were used to determine the relative magnitude of the training effects. Effects <0.41 represented a small ES, 0.41–0.70 a moderate ES, and >0.70 a large ES (3). The chances (% and qualitative) that the true value of the statistic (percent change in variable of interest) was practically or mechanistically positive, trivial, or negative was also calculated using the spreadsheet (10). This approach using probability statistics allows the reader to make decisions around the use of feedback based on its predicted beneficial or harmful effects in addition to statistical significance. Statistical power was calculated for each outcome variable based on an alpha level of 0.05 and difference in means and *SDs* between groups. An alpha level of 0.05 was also used for statistical significance. Confidence intervals (90%) and *p* values were also presented where appropriate.

RESULTS

The change in horizontal jump and 30-m sprint time were the only statistically significant differences between training groups ($p = 0.01$ and 0.0008, respectively). The mean ($\pm SD$) results and percent change of the performance test for the feedback and non-feedback conditions can be observed in Table 2. These show that for all tests the feedback condition produced larger percent changes in means (0.9–4.6% vs. -0.3 to 2.8%). With regards to practical significance, the chance that these changes were practically beneficial or trivial and the ESs are reported in Table 3. The probabilities that the use of feedback during squat jump training was beneficial were 45% for vertical jump performance, 65% for 10-m sprint performance, 49% for 20-m sprint performance, 83% for horizontal jump performance, and 99% for 30-m sprint performance. The relative magnitude (ES) of the training effects for all performance tests were found to be

TABLE 3. Effect sizes and chances (% and qualitative) that the benefit of feedback during jumps squats is practically positive or trivial for vertical jump, horizontal jump, and 10-/20-/30-m sprints after 6 weeks of training.

	Vertical jump	Horizontal jump	10-m sprint	20-m sprint	30-m sprint
Effect size	0.18 (Small)	0.28 (Small)	−0.28 (Small)	−0.20 (Small)	−0.46 (Moderate)
Positive	45 (Possibly)	83 (Likely)	65 (Possibly)	49 (Possibly)	99 (Almost certainly)
Trivial	51	17	33	49	1
Power	0.131	0.851	0.791	0.860	1.000

small (0.18–0.28), except for the 30-m sprint performance, which was moderate (0.46).

The ICC was used as a measure of consistency of effort between days. The ICCs for the feedback condition (0.81–0.95) were larger than for the non-feedback condition (−0.52 to 0.14) suggesting that those in the feedback group maintained effort (i.e., average system velocity) to better effect than the non-feedback group.

DISCUSSION

The purpose of this study was to investigate the effect of instantaneous performance feedback (peak velocity) provided after each repetition of squat jump exercises over a 6-week training block on sport-specific performance tests. This contention is subsequently discussed.

In terms of the performance measures, an increase in vertical jump over the 6 weeks was observed in both feedback (4.6%) and non-feedback (2.8%) conditions. Although a greater improvement was seen with feedback, there was a 51% chance that this was trivial and 45% chance of being positive. Given that this performance test was very similar to the movement used in training (squat jump), it suggests that improvements were seen as a result of repetition of the movement regardless of the feedback conditions. These results are similar to improvements in vertical jump (3.7%) observed after 5 weeks of squat jump training using a 70% 1RM load without feedback (9). Even though the load moved was greater than that used in this study, it again appears that repetition of the squat jump movement will result in an increase in vertical jump height. Previous research has also shown that even a squat program without a dynamic component has a positive effect in increasing vertical jump (1). The authors suggested that the squat was conducive to enhancing neuromuscular efficiency, in turn allowing for excellent transfer to other biomechanically similar movements requiring lower body triple extension movements as seen in the vertical jump. Although increases in vertical jump were seen both with and without the use of feedback, the use of feedback was reported to have a 45% chance of a positive effect on performance and produced a small training effect (ES = 0.18). This would suggest that there is evidence to

support the use of feedback during training to enhance vertical jump performance.

A larger increase in performance with the use of feedback was also observed in the horizontal jump (2.6 vs. 0.5%). As suggested previously, it is thought that movements requiring a powerful thrust from hips and thighs can be improved through the prescription of a biomechanically similar movement during training (1). It would seem that this has occurred here where the use of squat jumps during training resulted in improvements in horizontal jump performance. Again there appears justification for the use of feedback within training to optimize performance improvements, because the use of feedback was reported have a 83% chance of having a positive effect on horizontal jump performance and a small training effect noted (ES = 0.28).

Improvements in sprinting speed were observed over 10-m (1.3%), 20-m (0.9%), and 30-m (1.4%) distances. Again these were larger than those observed from the feedback group (0.1, 0.1, and −0.3%, respectively). The results from the non-feedback group are in agreement with the findings of previous research using jumps without feedback. Loads of 70% 1RM (9) and 30% (19) were also reported not to have produced significant increases in speed, questioning the effectiveness of squat jumps, regardless of relative load, in eliciting speed improvements. Although there have been reports of improvements in 10 m (1.6%) and 20 m (0.9%) times after 8 weeks of squat jump training using a 30%1RM load without feedback (14), it should be noted the subjects were recreationally active, involved in some type of club-level activities, whereas the present subjects were professional athletes. The pretraining times for the current non-feedback subjects were considerably faster (1.79 vs. 1.91 seconds and 3.06 vs. 3.27seconds, respectively) suggesting the current athletes had less scope for improvement.

What is of significance in this study are the increases in speed observed through the use of feedback during training. Feedback was reported to have a 65 and 49% chance of having a positive effect on 10- and 20-m sprint performance, respectively, with small training effects (ES = −0.28 and −0.20, respectively). In addition, feedback was reported to have a 99% chance of having a positive effect on 30-m performance, with a moderate training effect (ES = 0.46). This

may be because of the use of feedback during training enabling a greater consistency in the peak velocity achieved during the squat jumps. It has been suggested that the actual velocity of training is a vital component of producing high velocities (14). In addition, peak velocity during traditional squats has been shown to be significantly correlated to sprint time ($r = 0.40$, $p = 0.029$) (17). Similarly, it has also been suggested that exercises with a greater rate of force development (RFD) lead to greater improvements in sprinting (18), and although RFD was not measured in this study, consistently higher peak bar velocities were seen with feedback. Therefore, it would appear that optimizing the training session through the use of feedback leads to increases in sprint performance that may not have been realized using traditional training strategies.

Previous work by the authors has shown that the provision of feedback adds consistency to the performance of squat jumps (16) and increases peak velocity of squat jumps (15). It was suggested that these benefits may be transferred to movement or sport-specific tasks if applied over a training phase. With regards to the motivational aspects of feedback, it seems that the feedback condition resulted in consistency of effort and performance throughout the program as highlighted by the reported ICC values. The feedback condition ICCs ranged from 0.81 to 0.95, whereas the non-feedback condition ICCs ranged from -0.52 to 0.14 . Given that the ICCs relate to the reproducibility of the rank order of subjects on a subsequent training session, it appears that the feedback conditions enabled subjects to perform consistently in relation to the other subjects, whereas the non-feedback subjects varied greatly in their performance from session to session.

A number of limitations need to be acknowledged before the concluding remarks. First the sample size in each group was relatively small, but this represented all the professional players in the region. The aim was to use well-trained players because it is much more difficult to elicit adaptation and performance enhancement in well-trained athletes. As a result of the small sample size, the probability that the findings were practically significant were calculated. To many practitioners, such a statistic is invaluable, given that some results may not be statistically significant, but there may be a high probability that the intervention is practically or clinically beneficial as was the case for the 10-m sprint. That is, even though there was no significant difference between feedback and non-feedback conditions, there was a 65% probability that the use of feedback was beneficial to 10-m sprint performance. Given those odds, most practitioners would choose to use feedback even though not statistically significant.

A final limitation was the duration of the training study, that is, 6 weeks. Longer exposure to the intervention may have resulted in larger training effects. However, given that most training cycles are of 4- to 6-week durations, the duration of this study seems to have face or logical validity. Once more the results of this study (i.e., ~ 1 – 5% changes in the

performance measures of the feedback group) are noteworthy, given the duration of the intervention and training status of the subjects.

PRACTICAL APPLICATIONS

Of particular interest to the strength and conditioning practitioner is the observation that the provision of feedback on a single exercise (squat jump) during a resistance strength training program resulted in an improvement in the performance of movement and sport-specific tests. Given that athletes were also able to improve performance over a 6-week training program, it would seem intuitive to monitor multiple exercises of each training session and provide feedback, which should provide greater potential for adaptation and larger training effects. The use of such monitoring and feedback technologies may be further used through the ability to set training performance targets, such as maximum velocity and number of repetitions or sets completed above a predetermined performance threshold. This has the potential to eliminate the performance of repetitions that may be contributing to fatigue without providing a positive training effect, for example, power training. In addition, this may prove to be very motivational when fatigue sets in and creating competition between athletes in the training environment.

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